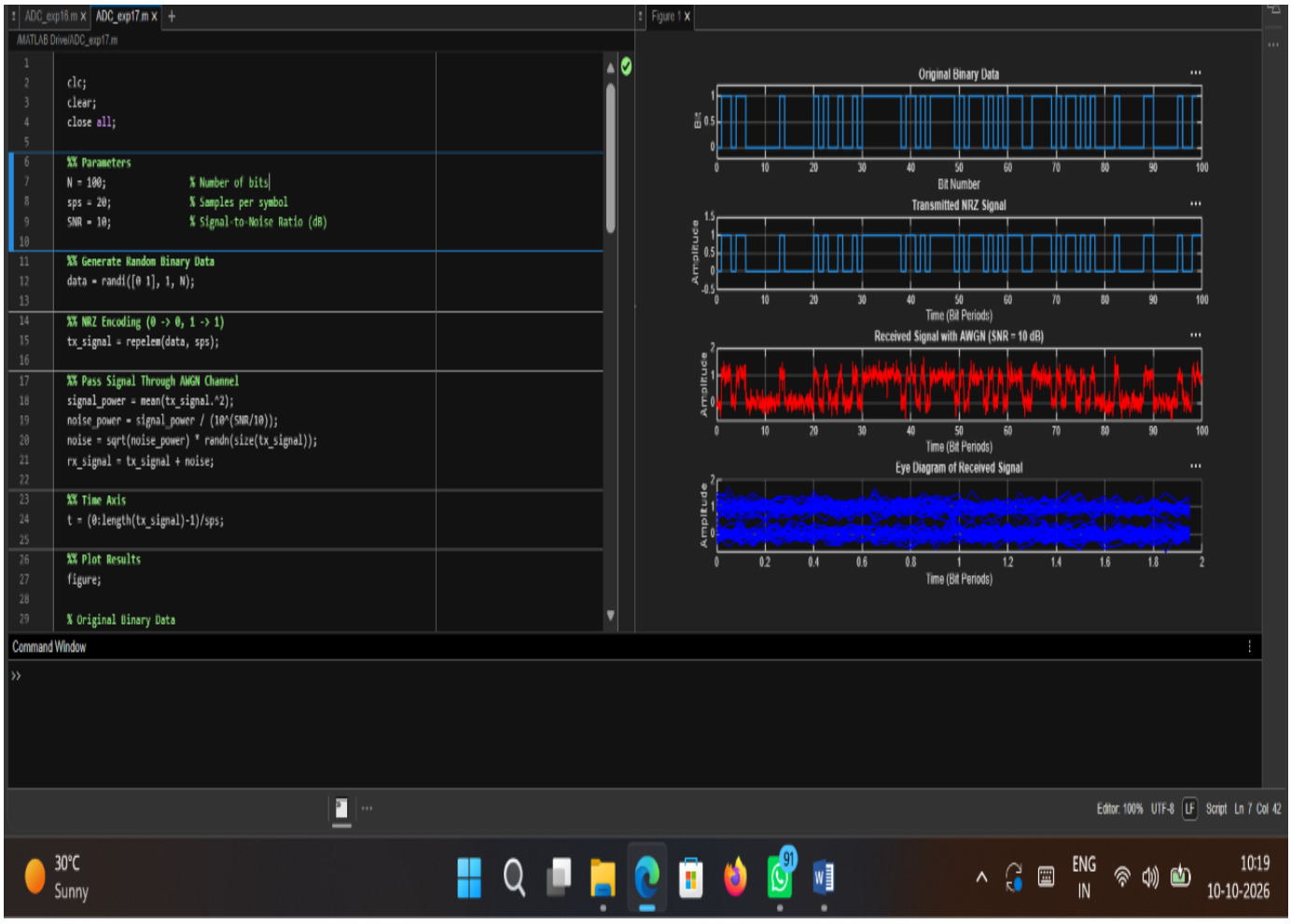
Ex.No:17 PERFORMANCE EVALUATION OF DIGITAL COMMUNICATION SYSTEMS USING EYE DIAGRAM ANALYSIS UNDER ADDITIVE WHITE GAUSSIAN NOISE (AWGN) USING MATLAB

Objective 1:



Code:

```
clc;
clear;
close all;

%% Parameters
N = 100;                    % Number of bits
sps = 20;                   % Samples per symbol
SNR = 10;                   % Signal-to-Noise Ratio (dB)

%% Generate Random Binary Data
data = randi([0 1], 1, N);
```

```
%% NRZ Encoding (0 -> 0, 1 -> 1)
tx_signal = repelem(data, sps);

%% Pass Signal Through AWGN Channel
signal_power = mean(tx_signal.^2);
noise_power = signal_power / (10^(SNR/10));
noise = sqrt(noise_power) * randn(size(tx_signal));
rx_signal = tx_signal + noise;

%% Time Axis
t = (0:length(tx_signal)-1)/sps;

%% Plot Results
figure;

% Original Binary Data
subplot(4,1,1);
stairs(0:N-1, data, 'LineWidth', 1.5);
grid on;
title('Original Binary Data');
xlabel('Bit Number');
ylabel('Bit');
ylim([-0.2 1.2]);

% Transmitted NRZ Signal
subplot(4,1,2);
plot(t, tx_signal, 'LineWidth', 1.5);
grid on;
title('Transmitted NRZ Signal');
xlabel('Time (Bit Periods)');
ylabel('Amplitude');
ylim([-0.5 1.5]);

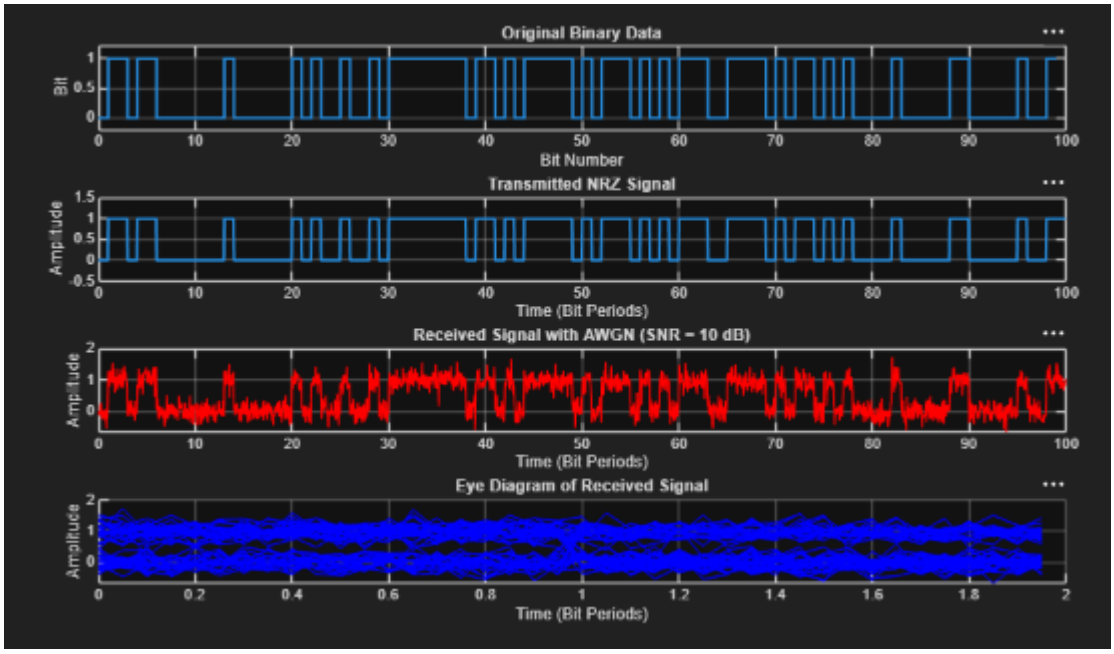
% Received Signal After AWGN
subplot(4,1,3);
plot(t, rx_signal, 'r');
grid on;
title(['Received Signal with AWGN (SNR = ', num2str(SNR), ' dB)']);
xlabel('Time (Bit Periods)');
ylabel('Amplitude');

% Eye Diagram (manual plotting; no eyediagram function required)
subplot(4,1,4);
hold on;

for k = 1:2:(N-2)
    idx = (k-1)*sps + 1;
    segment = rx_signal(idx:idx+2*sps-1);
    time_segment = (0:2*sps-1)/sps;
    plot(time_segment, segment, 'b');
end

grid on;
title('Eye Diagram of Received Signal');
xlabel('Time (Bit Periods)');
ylabel('Amplitude');
xlim([0 2]);
hold off;
```

Output:


Original Binary Data
Bit
Bit Number
Transmitted NRZ Signal
Amplitude
Time (Bit Periods)
Received Signal with AWGN (SNR = 10 dB)
Amplitude
Time (Bit Periods)
Eye Diagram of Received Signal
Amplitude
Time (Bit Periods)